

M18 PROJECT HANDBOOK

How to build evidence-backed portfolio projects

THE PROJECT LEARNING LOOP

- Orient before building.
- Create learner-owned work.
- Validate independently.
- Preserve failures and changed retries.
- Separate static from runtime truth.
- Route M18 -> M19 -> G07.

BEGINNER-FIRST PROFESSIONAL INSTITUTE REBUILD

Build -> prove -> break -> repair -> rerun -> defend. No solved portfolio project is supplied.

This book is your teacher

Purpose

Use this handbook before the four project books. It teaches how to work like a learner-engineer: requirements first, real artifacts, independent validation, failures preserved, runtime truth honest, and evidence easy to review.

Project loop

Brief -> source profile -> architecture -> build -> validate -> changed/failure test -> repair if needed -> clean rerun -> documentation -> defense.

Evidence before confidence

Evidence contract

Every phase evidence JSON records artifact paths, expected invariant, actual result, independent control, changed/failure test, runtime truth, explain-back, unresolved critical failure, and next action.

Why

A project screenshot can look convincing while the result is wrong. Evidence connects claims to reproducible checks.

How to use incomplete starters

Starter rule

A TODO/NotImplementedError/where 1=0/green placeholder test is a deliberate starting state, not a defect to "fix" by supplying a complete course answer.

Your job

Replace one incomplete portion only when its canonical phase asks you to. Keep commits/evidence phase-scoped so you can explain what changed.

Validation ladder

Before run

Write expected invariant/result shape first.

After run

Compare actual vs expected; use one independent control.

Change/break

Inject changed input, duplicate/fan-out, partial failure, skew/schema change, or other phase-matched fault.

Recover

Preserve first failure -> root cause -> smallest repair -> fresh retry -> rerun previous critical controls.

Runtime truth vocabulary

EXECUTED

You ran it in the claimed runtime and saved command/run/output evidence.

STATIC / SIMULATED

You inspected code, a local fixture, design, or deterministic control but did not execute the real external runtime.

NOT EXECUTED

Runtime/service/account/tool was unavailable or intentionally deferred. State this explicitly.

Portfolio publication standard

Show

Problem, architecture, source/grain, critical controls, one failure/recovery, clean reproduction, limitations/runtime truth, and evidence index.

Do not expose

Secrets, PII, protected answer/control files, private credentials, or fabricated cloud screenshots/logs.

Technical defense standard

90-second answer

Business goal -> source/grain -> architecture -> strongest correctness control -> failure/recovery -> biggest trade-off -> runtime truth -> revisit trigger.

Reviewer challenge

Be ready for: Why this tool? What breaks? How do you know totals are right? What happens on rerun? What did you not execute? What would you redesign at 10x?

Canonical route

Sequence

P01 -> P02 -> P03 -> CAP -> M19 -> G07.

Do not skip

CAP-12 does NOT jump directly to G07. M19 interview/technical-English/job-launch work comes first, then the final G07 practical.